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(54) **PROCESS FOR THE PRODUCTION OF
TETRAAMINOBIPHENOL MACROCYCLIC
LIGANDS; AND NOVEL
TETRAAMINOBIPHENOL MACROCYCLIC
LIGANDS**

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application No. PCT/GB2018/050547 on Mar. 2,
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CPC **C07D 257/10** (2013.01); **C07D 487/06**
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(58) **Field of Classification Search**
CPC C07D 257/10; C07D 487/06
See application file for complete search history.

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(57) **ABSTRACT**

A process for preparing a tetra-substituted aminobiphenol
macrocylic ligand having the structure (I), including the
step of treating a precursor compound having the structure
(II) with a compound having the structure R6-L where L is
a leaving group (hereafter compound (III)) in the presence of
a base.

18 Claims, No Drawings

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PROCESS FOR THE PRODUCTION OF TETRAAMINOBIPHENOL MACROCYCLIC LIGANDS; AND NOVEL TETRAAMINOBIPHENOL MACROCYCLIC LIGANDS

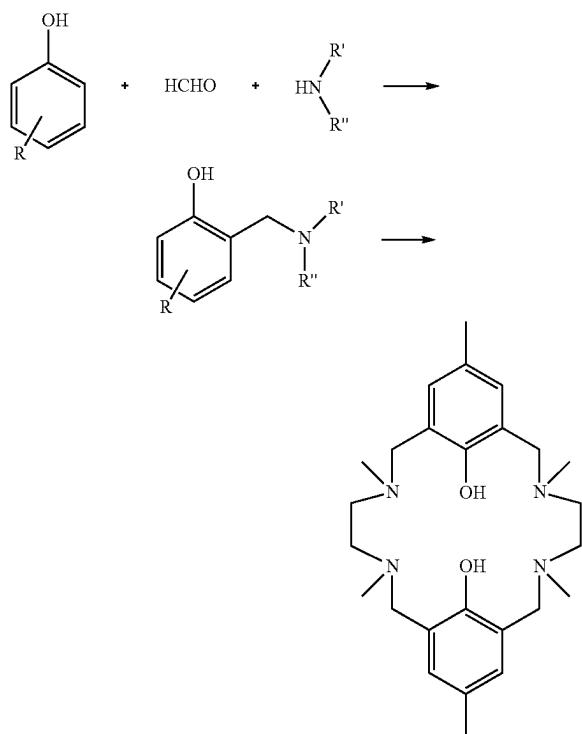
TECHNICAL FIELD AND BACKGROUND

The present invention relates to an improved process for the production of tetraaminobiphenol macrocyclic ligands, in particular, tetraaminobiphenol macrocyclic ligands for use in the production of bimetallic catalysts employed in copolymerization reactions.

The invention relates in particular to an improved process for the production of tetraaminobiphenol macrocyclic ligands in which each of the amino moieties of the macrocyclic ligand is substituted, for example alkylated or allylated.

The invention further relates to novel tetraaminobiphenol macrocyclic ligands.

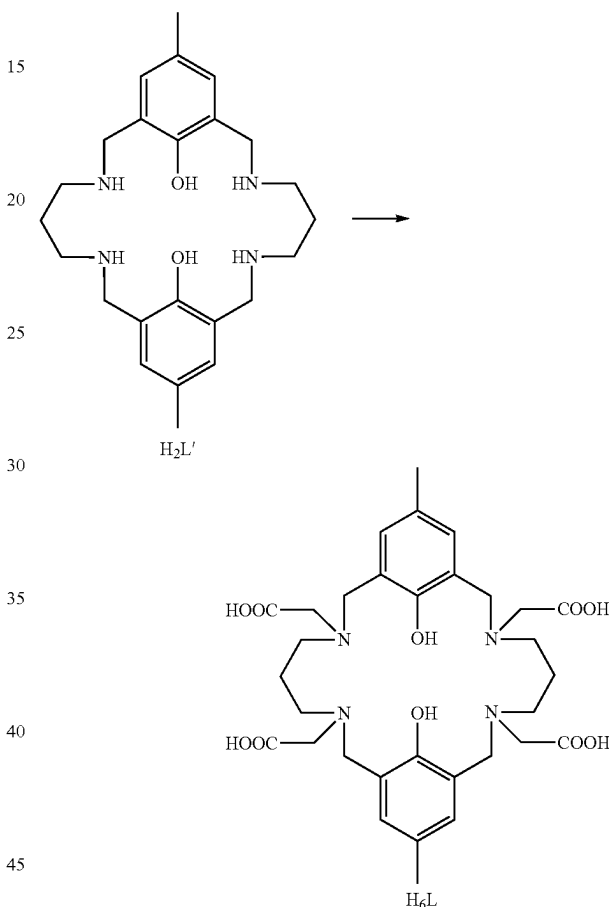
It is disclosed in Hodgkin, Polymer Science: Part A: Polymer Chemistry Edition, Vol. 24, 3117-3127 (1986), that a Mannich reaction between p-cresol, and N,N'-dimethylethylenediamine and formaldehyde yielded a tetramethylaminobiphenol macrocyclic ligand. The process is as shown in outline below.



In this method there is no alkylation step carried out on the macrocyclic ligand and therefore no risk of alkylating the phenol moieties. However, the process has disadvantages. The macrocycle is formed in low yields as a by-product to the linear polymeric products that were the target of Hodgkin et al. There is no indication that any other macrocyclic product could be made; but if this method were possible for different substituents of the amine groups it would be necessary to run a new Mannich reaction for each macrocyclic ligand.

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Direct alkylation of a tetraaminobiphenol macrocyclic ligand is described in Chen et al., CrystEngComm, 2013, 15, 5168-5178. In this reference it is stated that a tetraaminobiphenol macrocyclic ligand (3 mmol) was reacted with ethylchloroacetate (6 mmol) and potassium carbonate (24 mmol) in dimethylformamide under reflux for 20 hours. Following filtration and evaporation the residue was hydrolysed with sodium hydroxide to give the desired tetracarboxylic acid ligand. The reaction scheme is shown in outline below:



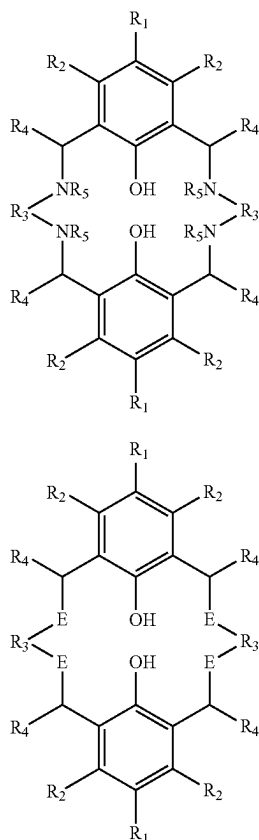
It is stated in the Chen reference that the tetracarboxyaminobiphenol ligand H₆L was obtained in 65% yield. However this yield is not possible: only two equivalents of ethylchloroacetate were used, per one equivalent of the tetraaminobiphenol macrocyclic ligand H₂L'. It will be appreciated that the tetraaminobiphenol macrocyclic ligand of interest in the Chen reference has four N-sites and two phenolic moieties, all of which are potential sites for reaction. Thus the theoretical maximum of the yield of the tetracarboxyaminobiphenol ligand is 50%; and this would assume no other products; for example no mono-carboxy ligand, no dicarboxyaminobiphenol ligand, no tri-carboxyaminobiphenol ligand, and no alkylated phenol ligands of any sort. It may be noted here that alkylation of phenol moieties under basic conditions is commonplace and is described in chemistry textbook. One literature example (from many) is *Facile Synthesis of Alkyl Phenyl Ethers Using Cesium Carbonate*, Lee et al, Synthetic Communications, 25(9), 13671370 (1995) which describes a "highly efficient" alkylation

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method of phenols using alkyl halides/cesium carbonate/ acetonitrile. For Chen to choose to use such a low amount of the alkylating agent—two mole equivalents when the target product has 4 moles of the carboxylate substituent— may indicate a desire to avoid competing reactions, such as alkylation of the phenol groups.

DETAILED DESCRIPTION

In accordance with a first aspect of the present invention there is provided a process for preparing a tetra-substituted aminobiphenol macrocyclic ligand having the structure (I), comprising the step of treating a precursor compound having the structure (II) with a compound having the structure R_6 -L where L represents a leaving group (hereinafter compound (III)) in the presence of a base;



wherein R₁ and R₂ are independently selected from hydrogen, halide, a nitro group, a nitrile group, an imine group, —NCR₁₃R₁₄, an amine, an ether group —OR₁₅ or —R₁₆OR₁₇, an ester group —OC(O)R₁₀ or —C(O)OR₁₀, an amido group —NR₉C(O)R₉ or —C(O)—NR₉ (R₉), —COOH, —C(O)R₁₅, —OP(O)(OR₁₈)(OR₁₉), —P(O)R₂₀R₂₁, a silyl group, a silyl ether group, a sulfoxide group, a sulfonyl group, a sulfinate group or an acetylide group or an optionally substituted alkyl, alkenyl, alkynyl, haloalkyl, aryl, heteroaryl, alkoxy, aryloxy, alkylthio, arylthio, alicyclic or heteroalicyclic group;

R₃ is independently selected from optionally substituted alkylene, alkenylene, alkynylene, heteroalkylene, heteroalkenylene, heteroalkynylene, arylen, het-

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eroarylene or cycloalkylene, wherein alkylene, alkenylene, alkynylene, heteroalkylene, heteroalkenylene and heteroalkynylene, in each case optionally interrupted by aryl, heteroaryl, alicyclic or heteroalicyclic; R₄ is independently selected from hydrogen, or optionally substituted aliphatic, heteroaliphatic, alicyclic, heteroalicyclic, aryl, heteroaryl, alkylheteroaryl or alkylaryl;

R₅ is independently selected from hydrogen, optionally substituted aliphatic, heteroaliphatic, alicyclic, alkanolate, arylate, carboxyl, heteroalicyclic, aryl, heteroaryl, alkylheteroaryl or alkylaryl,

or two R₅ species may together be selected from optionally substituted alkylene, alkenylene or alkynylene, bonded to two different N groups of the compound of structure (II),

with the proviso that at least one of the species R₅ is hydrogen;

and E is independently selected from NR₅ and N₆, with the proviso that at least one of the species E is NR₆; wherein R₆ is independently selected from optionally substituted aliphatic, alicyclic, heteroalicyclic, aryl, heteroaryl, ether, polyether, or optionally substituted alkylaryl or alkylheteroaryl; and

wherein R₉, R₁₀, R₁₃, R₁₄, R₁₈, R₁₉, R₂₀ and R₂₁ are independently selected from hydrogen or an optionally substituted aliphatic, heteroaliphatic, alicyclic, heteroalicyclic, aryl or heteroaryl group and R₁₅, R₁₆ and R₁₇ are independently selected from an optionally substituted aliphatic, heteroaliphatic, alicyclic, heteroalicyclic, aryl or heteroaryl group;

wherein the molar ratio of compound of structure (III) to the number of NH sites in the compound of structure (II) is at least 0.6.

Preferably the molar ratio of the compound having the structure (III) to species R₅ which are hydrogen in the compound having the macrocycle (II)—i.e. the number of NH sites in the macrocycle—is at least 0.8.

Preferably the molar ratio of the compound having the structure (III) to the number of NH sites in the macrocycle (II) is at least 1.

In especially preferred embodiments the molar ratio of the compound having the structure (III) to the number of NH sites in the macrocycle (II) is at least 1.1.

In a preferred process of the present invention the molar ratio of the compound having the structure (III) to the number of NH sites in the macrocycle (II) is not greater than 2; preferably not greater than 1.6; preferably not greater than 1.4.

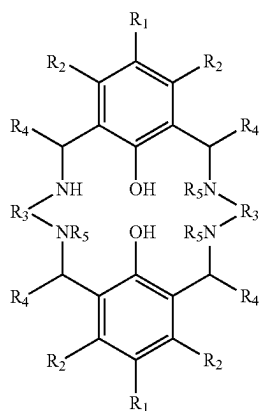
The number of NH sites in the macrocycle (II)' may alternatively be referred to as 'the species R₅ which are hydrogen in compound (II)'.

In preferred embodiments the molar ratio of the compound of structure (III) to the compound of structure (II) is at least 2.2, preferably at least 2.4.

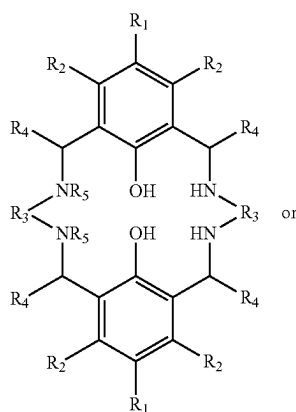
One sub-group of compounds having the structure (IIa) shown below (and conforming to the more general structure (II)) has a single group R₅ which is hydrogen and the remaining three groups R₅ which are any of the groups mentioned above, but not hydrogen. In this subgroup the molar ratio of the compound having the structure (III) to the compound of structure (IIa) is at least 0.6, preferably at least 0.8, more preferably at least 1, and in especially preferred embodiments at least 1.1. In the sub-group of compounds having the structure (IIa) the molar ratio of the compound having the structure (III) to the compound of structure (IIa)

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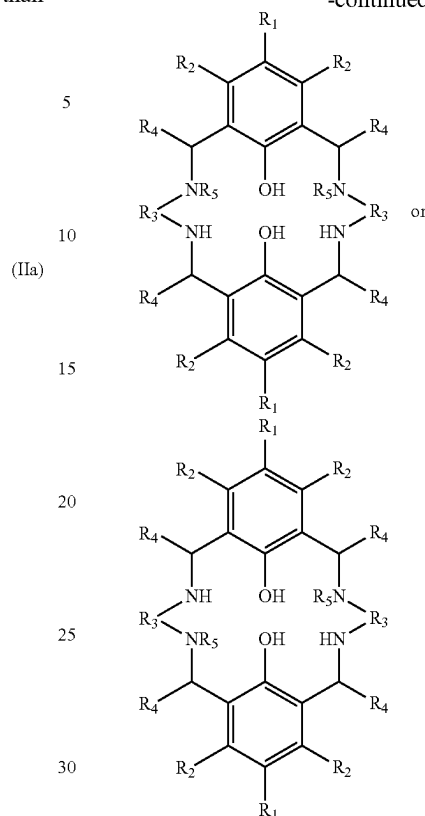
is preferably not greater than 2; preferably not greater than 1.6; preferably not greater than 1.4.



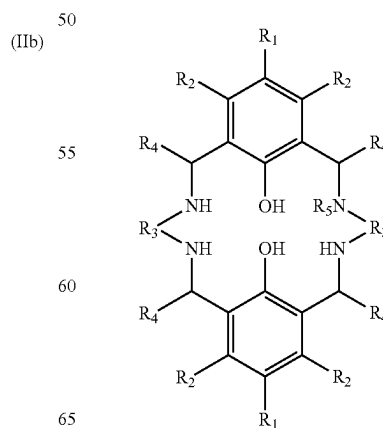
One sub-group of compounds having the structure (IIb) shown below (and conforming to the more general structure (II)) has two groups R_5 which are hydrogen and the remaining two groups R_5 which are any of the groups mentioned above, but not hydrogen. In this subgroup the molar ratio of the compound having the structure (III) to the compound of structure (IIb) is at least 1.2, preferably at least 1.6, more preferably at least 2, and in especially preferred embodiments at least 2.2. In the sub-group of compounds having the structure (IIb) the molar ratio of the compound having the structure (III) to the compound of structure (IIb) is preferably not greater than 4; preferably not greater than 3.2; preferably not greater than 2.8.

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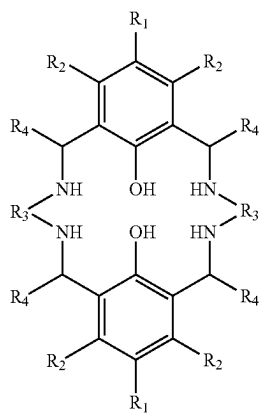


One sub-group of compounds having the structure (IIc) shown below (and conforming to the more general structure (II)) has three group R_5 which are hydrogen and the remaining single group R_5 which is any of the groups mentioned above, but not hydrogen. In this subgroup the molar ratio of the compound having the structure (III) to the compound of structure (IIc) is at least 1.8, preferably at least 2.4, more preferably at least 3, and in especially preferred embodiments at least 3.3. In the sub-group of compounds having the structure (IIc) the molar ratio of the compound having the structure (III) to the compound of structure (IIc) is preferably not greater than 6; preferably not greater than 4.8; preferably not greater than 4.2.



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One preferred sub-group of compounds having the structure (II_d) shown below and (conforming to the more general structure (II)) has all four groups R₅ which are hydrogen. In this subgroup the molar ratio of the compound having the structure (III) to the compound of structure (II_d) is at least 2.4, preferably at least 3.2, preferably at least 3.6, more preferably at least 4, more preferably at least 4.2, and in especially preferred embodiments at least 4.4. In the subgroup of compounds having the structure (II_d) the molar ratio of the compound having the structure (III) to the compound of structure (II_d) is preferably not greater than 8; preferably not greater than 6.4; preferably not greater than 5.6.



The process of the present invention uses a significantly higher molar ratio of compound (III) to species R₅ which are hydrogen in compound (II), compared with the prior art of Chen, mentioned above. In Chen's publication the molar ratio of the chloroethylacetate to the four alkylatable NH groups on the ligand H₂L' is 0.5.

Surprisingly, the present inventors have found that very high yields of the target substituted aminobiphenol ligands of formula I can be obtained, despite the presence of the phenol sites at which reaction might be expected, and use of a high molar ratio of reactant (III). To the inventors' surprise, the use of such a high mole ratio of the reactant (III) does not appear to promote competing reactions at the phenolic —OH sites. Rather, it produces the compound of formula (I) in high yield and at high selectivity.

Each of the occurrences of the groups R₁ and R₂ may be the same or different. Preferably, R₁ and R₂ are independently selected from hydrogen, halide, amino, nitro, sulfoxide, sulfonyl, sulfinate, silyl, silyl ether and an optionally substituted alkyl, alkenyl, aryl, heteroaryl, heteroalicyclic, alkoxy, aryloxy or alkylthio. Preferably, each occurrence of R₂ is the same, and is hydrogen or alkyl, for example methyl. Preferably R₂ is hydrogen.

Even more preferably, R₂ is alkyl or, especially, hydrogen, and R₁ is independently selected from hydrogen, halide, amino, nitro, sulfoxide, sulfonyl, sulfinate, silyl, silyl ether and optionally substituted alkyl, alkenyl, aryl, heteroaryl, heteroalicyclic, alkoxy, aryloxy, alkylthio or arylthio, such as hydrogen, C₁₋₆alkyl (e.g. haloalkyl), alkoxy, aryl, halide, nitro, sulfonyl, silyl and alkylthio, for example t-butyl, n-butyl, i-propyl, methyl, piperidinyl, methoxy, hexyl methyl ether, —SCH₃, —S(C₆H₅), H, nitro, trimethylsilyl, methylsulfonyl (—SO₂CH₃), triethylsilyl, halogen or phenyl.

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Each occurrence of R₁ can be the same or different, and R₁ and R₂ can be the same or different. Preferably each occurrence of R₁ is the same. Preferably each occurrence of R₂ is the same. When R₁ and R₂ are the same, preferably each occurrence of R₁ and R₂ is hydrogen or methyl. Preferably, each occurrence of R₁ is the same, and each occurrence of R₂ is the same, and R₁ is different to R₂.

Preferably both occurrences of R₁ are the same, and are selected from hydrogen, halide, amino, nitro, sulfoxide, sulfonyl, sulfinate, silyl, silyl ether and an optionally substituted alkyl, alkenyl, aryl, heteroaryl, heteroalicyclic, alkoxy, aryloxy, or alkylthio. More preferably both occurrences of R₁ are the same, and are selected from halide, sulfoxide, silyl, and an optionally substituted alkyl, heteroaryl or alkoxy. Still more preferably both occurrences of R₁ are the same, and are selected from H, alkyl, aryl, alkoxy, trialkylsilyl such as triethylsilyl, or halide. More preferably still both occurrences of R₁ are the same, and are selected from H, alkyl, phenyl, halide or trialkylsilyl. Most preferably, both occurrences of R₁ are the same, and are selected from H, methyl, ethyl, n-propyl, i-propyl n-butyl, t-butyl, t-amyl, t-octyl, methylthio, methoxy and triethylsilyl.

It will be appreciated that the group R₃ can be the divalent alkylene, alkenylene, alkynylene, heteroalkylene, heteroalkenylene or heteroalkynylene group which may optionally be interrupted by an aryl, heteroaryl, alicyclic or heteroalicyclic group, or may be a divalent arylene or cycloalkylene group which acts as a bridging group between two nitrogen centres in the macrocycle of formula (I). Thus, where R₃ is an alkylene group, such as 2,2-dimethylpropane-1,3-diyl, the R₃ group has the structure —CH₂—C(CH₃)₂—CH₂—. The definitions of the alkyl, aryl, cycloalkyl etc groups set out herein therefore also relate respectively to the divalent alkylene, arylene, cycloalkylene etc groups set out for R₃, and may also be optionally substituted. Exemplary options for R₃ include ethane-1,2-diyl, 2,2-fluoropropane-1,3-diyl, 2,2-dimethylpropane-1,3-diyl, propane-1,3-diyl, butane-1,4-diyl, phenylene, cyclohexane-1,4-diyl, cyclohexane-1,2-diyl or biphenylene. When R₃ is cyclohexane-1,4-diyl or cyclohexane-1,2-diyl, it can be the racemic, RR— or SS— forms.

R₃ can be independently selected from substituted or unsubstituted alkylene, alkenylene, alkynylene, heteroalkylene, heteroalkenylene or heteroalkynylene, arylene or cycloalkylene. Preferably, R₃ is selected from substituted or unsubstituted alkylene, cycloalkylene, alkenylene, heteroalkylene and arylene. More preferably, R₃ is selected from —CH₂C(CH₃)₂CH₂—, —CH₂CH₂CH₂—, —CH₂CH(CH₃)CH₂—, —CH₂C(CH₂C₆H₅)₂CH₂—, —(C₆H₄)—, —CH₂CH₂—, —CH₂—CH₂CH₂CH₂—, —CH₂CH₂N(CH₃)CH₂CH₂—, —(C₆H₁₀)— or —CH₂CH₂CH(C₂H₅)—. Still more preferably R₃ is selected from —CH₂C(CH₃)₂CH₂—, —CH₂CH₂—, —CH₂CH₂CH₂—, —CH₂CH(CH₃)CH₂—, —CH₂C(CH₂C₆H₅)₂CH₂—, —CH₂CH₂CH(C₂H₅)—, —CH₂CH₂CH₂CH₂—. More preferably still, R₃ is selected from —CH₂C(CH₃)₂CH₂—, CH₂CH₂CH₂—, —CH₂CH(CH₃)CH₂— and —CH₂C(C₂H₅)₂CH₂—.

R₃ can be independently selected from substituted or unsubstituted alkylenes and substituted or unsubstituted arylenes, preferably substituted or unsubstituted propylenes, such as propane-1,3-diyl and 2,2-dimethylpropane-1,3-diyl, and substituted or unsubstituted phenylene or biphenylene. Preferably both occurrences of R₃ are the same. Even more preferably R₃ is a substituted propane-1,3-diyl, such as 2,2-di(alkyl)propane-1,3-diyl, especially 2,2-dimethylpropane-1,3-diyl.

Preferably, each R₄ is independently selected from hydrogen, and optionally substituted aliphatic or aryl. More pref-

erably, each R_4 is independently selected from hydrogen, and optionally substituted alkyl or aryl. Even more preferably, each R_4 is the same, and is selected from hydrogen, and optionally substituted alkyl or aryl. Exemplary R_4 groups include hydrogen, methyl, ethyl, n-propyl, n-butyl, phenyl and trifluoromethyl, preferably hydrogen, methyl or trifluoromethyl. Even more preferably, each R_4 is hydrogen.

In preferred combinations of the R_4 group and R_1 group, R_1 is selected from H, methyl, ethyl, n-propyl, n-butyl, t-butyl, t-octyl, Cl, Br, F, nitro, trimethylsilyl, triethylsilyl, methoxy or methylthio and R_4 is selected from H, methyl, ethyl, n-propyl, phenyl, trifluoromethyl.

Preferably R_5 is independently selected from hydrogen, optionally substituted aliphatic, heteroaliphatic, alicyclic, alkanoate, arylate, carboxyl, heteroalicyclic, aryl, heteroaryl, alkylheteroaryl or alkylaryl; or two R_5 groups may together be an optionally substituted alkylene group, for example a C(1-10alkylene) group optionally substituted by an optionally substituted aryl group. At least one species R_5 is hydrogen. In embodiments of the invention one, two, three or four species R_5 may be hydrogen, as shown in structures (IIa)-(IIc) above. Species NR_5 which are NH provide a site or sites for reaction of the compound of structure (III) and introduction of the species R_6 . Thus an NH species in the compound of structure (II) can provide the site for an NR_6 species in the compound of structure (I).

Preferably R_5 is independently selected from hydrogen, optionally substituted alkyl which is optionally interrupted by at least one N atom, or by at least one O atom, or by at least one S atom, optionally substituted alkylthio, alkylaryl, alkylheteroaryl, alkenyl, alkynyl, heteroalkenyl, heteroalkynyl, aryl, heteroaryl, alicyclic, heteroalicyclic, sulfonate, alkanoate ($-C(O)-OR_{10}$), carbonyl ($-C(O)-R_{10}$), ether or, polyether, arylate, or carboxyl. More preferred R_5 species include optionally substituted groups selected from alkyl (such as methyl, ethyl, propyl, butyl), alkylaryl, alkylheteroaryl (such as benzyl, substituted benzyl or methyl pyridine), alkenyl (such as allyl), alkynyl (such as propargyl), sulfonate (such as tosylate), alkanoate (such as $-C(O)-O-t-butyl$ (BOC), $-C(O)-O-benzyl$, $-C(O)-O-9-fluorenylmethyl$ (Fmoc)), and optionally substituted aryl-C(O) OR_{10} , alkylaryl-C(O) OR_{10} , alkyl-C(O)- OR_{10} (such as $-CH_2-C(O)-OMe$, $-CH_2CH_2-C(O)-OMe$) or alkyl-C $\equiv$ N (such as $-CH_2CH_2-C\equiv N$). Most preferred species R_5 include methyl, ethyl, propyl (preferably n-propyl), butyl (preferably n-butyl), allyl, propargyl, benzyl, 4-nitrobenzyl, $-CH_2CH_2-C(O)-OR_{10}$, $-CH_2CH_2-C\equiv N$, or $-C(1-4)alkylC(O)O-$, for example, acetate or propanoate, or optionally substituted alkylarylC(O)O- or arylC(O)O-, for example benzoate which is unsubstituted or ring-substituted by 1, 2 or 3 C(1-4)alkyl groups. It will be appreciated that some of the groups defined above will be associated with a cation. The cation will typically be a sodium, lithium or ammonium cation.

When there is a plurality of groups R_5 which are not hydrogen, such groups may be identical or they may all differ from each other.

In some embodiments two R_5 species are together selected from methylene $-CH_2-$, ethylene $-CH_2-CH_2-$, propylene $-CH_2-CH_2-CH_2-$ or phenylmethylene $-CH(Ph)-$, bonded to two different N groups of the compound of structure (II).

Preferably each R_6 is independently selected from optionally substituted aliphatic, alicyclic, heteroalicyclic, aryl, heteroaryl, ether, polyether, or optionally substituted alkylaryl or alkylheteroaryl. More preferably, each R_6 is independently selected from optionally substituted alkyl (such as

methyl, ethyl, propyl, butyl), alicyclic (such as cyclohexane), alkenyl (such as allyl), alkynyl (such as propargyl) or alkylaryl (such as benzyl, substituted benzyl, methylpyridine, ether (such as methoxymethyl), polyether (such as polyethylene glycol (PEG), polypropylene glycol (PPG)), Preferred R_6 groups include C(1-10)alkyl, preferably C(1-6)alkyl, more preferably C(1-4)alkyl, most preferably methyl, ethyl, n-propyl, n-butyl; or allyl or propargyl or benzyl.

In preferred combinations of the R_6 group and R_1 group, R_1 is selected from H, methyl, ethyl, n-propyl, n-butyl, t-butyl, t-octyl, Cl, Br, F, nitro, trimethylsilyl, triethylsilyl, methoxy or methylthio and R_6 is selected from methyl, ethyl, n-propyl, n-butyl, allyl and benzyl.

When there is a plurality of groups R_6 such groups are preferably identical.

Preferably each R_5 group is different from each R_6 group.

The compound having the structure (III) introduces groups R_6 to provide the compound of formula (I). The leaving group L of the compound having the structure (III) is preferably selected from halogen, for example: chloro, bromo or iodo; or an ether group (for example when the compound R_6-L is a trialkyloxonium compound); or a tertiary amine (for example when the compound R_6-L is a quaternary ammonium salt, for example a tetraalkyl ammonium salt; or a sulfonate group of formula $-O-SO_2-R_x$ where R_x is selected from optionally substituted aliphatic or aryl or alkaryl; for example an alkyl sulfonate, aryl sulfonate, halo sulfonate or trifluoroalkyl sulfonate, for example tosylate, mesylate, triflate, fluorosulfonate, nosylate or brosylate; or a group of formula $-O-SO_2-O-R_y$ or $-O-CO-O-R_y$ where R_y represents optionally substituted aliphatic or aryl or alkaryl, preferably an alkyl group, for example methyl and ethyl. Sulfonate compounds suitable for use in the present invention may be polymer-bound, for example on polystyrene. When the leaving group L is of formula $-O-SO_2-O-R_y$ or $-O-CO-O-R_y$ it is preferred that R_6 and R_y are identical. Preferably they are both C₍₁₋₄₎ alkyl groups, preferably both methyl or both ethyl.

A base for use in the process of the first aspect may be an inorganic base or an organic base. Preferred inorganic bases include metal compounds, for example of Group I and Group II metals. Preferred metal compounds are carbonates, hydrogen carbonates, alkanoates, hydroxides, silicates, phosphates and borates. Preferred cationic species thereof are Group I metals, most preferably sodium, potassium or cesium; Group II metals, preferably magnesium, calcium, strontium and barium; or ammonium. Preferred anionic species of the metal compounds are C(1-4) alkanoates (especially acetates), carbonates, and hydroxides. Preferred bases include sodium carbonate, potassium carbonate, cesium carbonate, magnesium carbonate, calcium carbonate, ammonium carbonate, sodium hydroxide, potassium hydroxide, magnesium hydroxide, calcium hydroxide, ammonium hydroxide, sodium C(1-10) alkanoates, preferably sodium C(1-4) alkanoates, especially sodium acetate, potassium alkanoates, preferably potassium C(1-4) alkanoates, especially potassium acetate, sodium metasilicate, potassium metasilicate, sodium borate, potassium borate, trisodium phosphate, disodium phosphate, monosodium phosphate, monopotassium phosphate, dipotassium phosphate and tripotassium phosphate. Especially preferred bases are sodium carbonate, potassium carbonate, sodium acetate, potassium acetate, mono-, di- or trisodium phosphate, mono-, di- or tripotassium phosphate, sodium hydroxide or potassium hydroxide.

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Preferred organic bases include tertiary amine bases, suitably carrying groups selected from alkyl and alkylene. Preferably tertiary alkylamine bases have from 1 to 3 amine groups and C(1-10) alkyl or C(1-10) alkylene groups in place of all N-hydrogen species. Preferred tertiary alkylamine bases are of formula $(C(1-10)alkyl)_3-N$, especially tripropylamine, triethylamine, tributylamine, tri-isobutylamine or diisopropylethylamine. Preferred organic bases may further include trialkanolamine bases, especially triethanolamine.

A preferred base for use in the process of the invention has a pKb in the range from -2 to 14 when in an aqueous solution. It will be understood that the process of the present invention need not employ aqueous conditions and therefore that the pKb of the base does not represent any measurable pKb under the reaction conditions. Rather it is used as a representation of the strength of the base and it is believed to provide a useful guide to bases which are suitable for use in the present invention. In preferred embodiments the pKb in the range at least 0 (zero) to less than 13.5. Preferably the base has a pKb of at least 0, preferably at least 0.5, preferably at least 1, more preferably at least 2, more preferably at least 3. Preferably it has a pKb of not greater than 13, preferably not greater than 12, more preferably not greater than 10.

The process of the present invention may take place in the presence of a solvent. However in embodiments of the invention the reactants described above, in particular the compound of structure (III) and/or the base, may function as the solvent, and no further solvent is employed.

When a solvent is employed (in addition to the reactant species), it may suitably be selected from substituted alkyl, optionally substituted alkanol (especially halogenated alkanol; suitable halogen atoms being chlorine, bromine and iodine), a cyclic ether, an aliphatic ether, a dialkylformamide, a ketone, an ester, a dialkyl sulfoxide, an alkyl nitrile optionally substituted aryl; or optionally substituted heteroaryl. Water may also be used in some embodiments. A single solvent or a mixture of solvents may be used.

Preferred solvents have up to 20 carbon atoms, preferably up to 12 carbon atoms, preferably up to 8 carbon atoms.

Preferred substituted alkyl groups include halo C(1-4) alkyl groups, preferably chloro C(1-4)alkyl groups, for example chloroform or dichloromethane.

Preferred alkanols and haloalkanols are methanol, ethanol, propanol, iso-propanol, iso-butanol, t-butyl alcohol, phenol, n-butanol and chlorocapryl alcohol. Especially preferred alkanols are methanol and ethanol. A particularly preferred alkanol is methanol.

Preferred cyclic ethers are optionally substituted cycloaliphatic compounds having from 5-8 ring atoms of which at least one ring atom is an oxygen atom. Preferred substituents are one or two C(1-4) alkyl groups. Examples of suitable cyclic ethers include tetrahydrofuran, C(1-4)alkyl-substituted tetrahydrofurans, for example 2-methyl tetrahydrofuran, dioxane and dioxolane.

Preferred aliphatic ethers include simple ethers, for example alkylene interrupted by one or more oxygen atoms, being for example ethers of formula $R-O-R$ where each R represents an alkyl group; or may be glycols, glycol ethers and dialkyl glycols. Preferred simple ethers have the formula $C(1-6)alkyl-O-C(1-6)alkyl$, for example dibutyl ether. Preferred glycols have the formula $HO-C(1-6)alkyl-OH$. Preferred glycol ethers have the formula $C(1-6)alkyl-O-C(1-6)alkyl-OH$. Preferred dialkyl glycols have the formula $C(1-6)alkyl-O-C(1-6)alkyl-O-C(1-6)alkyl$, for example dimethoxyethane.

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Preferred alkyl nitriles are C(1-10)alkyl nitriles, preferably C(1-4)alkyl nitriles, for example acetonitrile.

Preferred dialkylformamides are di-C(1-10)alkyl formamides, preferably C(1-4)alkyl formamides, for example dimethyl formamide.

Preferred ketones are diC(1-10)alkyl ketones, preferably diC(1-4)alkyl ketones, for example acetone.

Preferred esters are C(1-10)alkanoate esters, for example C(1-4)alkyl C(1-4) alkanoate esters, for example ethyl acetate.

Preferred dialkyl sulfoxides are di-C(1-10)alkyl sulfoxides, preferably di-C(1-4)alkyl sulfoxides, for example dimethyl sulfoxide.

Preferred optionally substituted aryl groups include benzene substituted by 1-3 C(1-4) alkyl groups. Especially preferred alkylbenzene solvents include toluene, xylene and ethylbenzene.

A preferred optionally substituted heteroaryl group is pyridine.

Preferably, the process of the present invention is carried out at a temperature in the range 0 to 100° C., more preferably between 20 to 80° C. Some preferred embodiments are carried out at ambient temperature. Other preferred embodiments are carried out at the reflux temperature of the reaction system.

In a preferred process of the present invention the molar ratio of the base to the precursor compound having the structure (II) is at least 1; preferably at least 2; in some embodiments it is preferably at least 4; preferably at least 5.

In a preferred process of the present invention the molar ratio of the base to the precursor compound having the structure (II) is not greater than 7.5; preferably not greater than 6.

Preferably the molar ratio of the base to the number of NH sites in the compound of structure (II) is from 1 to 1.9, preferably from 1.1 to 1.8, and most preferably from 1.25 to 1.75.

Biphenol ligand precursors having the structure (II) used in the present invention and having four NH groups in the macrocyclic ring are described in publically available sources; for example in WO2016/012785 and in Chen et al., CrystEngComm, 2013, 15, 5168-5178, described above.

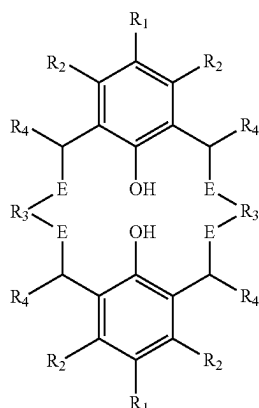
Biphenol ligand precursors having the structure (II) used in the present invention and having one, two or three NH groups in the macrocyclic ring (the other N atom having a substituent) may be prepared using protecting group chemistry in the absence of a base, as described in WO2016/012785. Alternative methods are described in Hodgkin, Polymer Science: Part A: Polymer Chemistry Edition, Vol. 24, 3117-3127 (1986), mentioned above.

Preferably the process of the present invention does not employ a protecting group protecting the phenol moieties. Surprisingly, it has been found that the process proceeds in high yield to the tetrasubstituted amino product of formula (I) even without such protection. In preferred embodiments of the invention yields of greater than 50% of the tetrasubstituted amino product of formula (I) are achieved, even without such protection. In preferred examples substantially higher yields have been obtained.

In the process of the invention the base may be dissolved in the reaction system. The base may be completely dissolved already at the start of the reaction. Alternatively the base may be heterogeneous, or may be partially dissolved, but able to accept protons as the reaction proceeds.

In accordance with a second aspect of the invention there is provided a compound of structure (Ie)

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wherein R_1 and R_2 are independently selected from hydrogen, halide, a nitro group, a nitrile group, an imine group, $\text{—NCR}_{13}\text{R}_{14}$, an amine, an ether group —OR_{15} or $\text{—R}_{16}\text{OR}_{17}$, an ester group —OC(O)R_{10} or —C(O)OR_{10} , an amido group $\text{—NR}_9\text{C(O)R}_9$ or —C(O)—NR_9 (R_9), —COOH , —C(O)R_{15} , $\text{—OP(O)(OR}_{18})(\text{OR}_{19})$, $\text{—P(O)R}_{20}\text{R}_{21}$, a silyl group, a silyl ether group, a sulfoxide group, a sulfonyl group, a sulfinate group or an acetylide group or an optionally substituted alkyl, alkenyl, alkynyl, haloalkyl, aryl, heteroaryl, alkoxy, aryloxy, alkylthio, arylthio, alicyclic or heteroalicyclic group;

R_3 is independently selected from optionally substituted alkylene, alkenylene, alkynylene, heteroalkylene, heteroalkenylene, heteroalkynylene, arylene, heteroarylene or cycloalkylene, wherein alkylene, alkenylene, alkynylene, heteroalkylene, heteroalkenylene and heteroalkynylene, in each case optionally interrupted by aryl, heteroaryl, alicyclic or heteroalicyclic; R_4 is independently selected from hydrogen, or optionally substituted aliphatic, heteroaliphatic, alicyclic, heteroalicyclic, aryl, heteroaryl, alkylheteroaryl or alkylaryl;

and E is independently selected from NR'_5 and NR_6 ,

wherein R'_5 is independently selected from hydrogen, optionally substituted aliphatic, heteroaliphatic, alicyclic, alkanoate, arylate, carboxyl, heteroalicyclic, aryl, heteroaryl, alkylheteroaryl or alkylaryl;

or two R_5 species may together be selected from optionally substituted alkylene, alkenylene or alkynylene, bonded to two different N groups of the compound of structure (II);

wherein R_6 is independently selected from optionally substituted aliphatic, alicyclic, heteroalicyclic, aryl, heteroaryl, ether, polyether, or optionally substituted alkylaryl or alkylheteroaryl;

with the first proviso that at least one of the species E conforms to the definition given for NR_6 ;

and with the second proviso that the groups R'_5 and R_6 are not all identical with each other.

wherein R_9 , R_{10} , R_{13} , R_{14} , R_{18} , R_{19} , R_{20} and R_{21} are independently selected from hydrogen or an optionally substituted aliphatic, heteroaliphatic, alicyclic, heteroalicyclic, aryl or heteroaryl group and R_{15} , R_{16} and R_{17} are independently selected from an optionally substituted aliphatic, heteroaliphatic, alicyclic, heteroalicyclic, aryl or heteroaryl group.

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Thus these compounds of structure (Ie) are asymmetrical as regards their groups R'_5 and R_6 .

In this aspect of the invention R_1 , R_2 , R_3 , R_4 , E , R_6 , R_9 , R_{10} , R_{13} , R_{14} , R_{18} , R_{19} , R_{20} and R_{21} may be as defined in any statement or claim contained in this specification. R'_5 may be as defined about for R_5 , except that it cannot be hydrogen.

It is the process of the first aspect that enables these compounds of structure (Ie). In the process the NH group or groups in the macrocycle of structure (II) are replaced on introduction of group(s) R_6 , whilst N-substituent group or groups already on the macrocycle (II) remain. The groups on the macrocycle, designated R'_5 and R_6 in the definition of the compounds (Ie), can be selected to be non-identical. Thus asymmetrical compounds of structure (Ie) may be provided.

Definitions

For the purpose of the present invention, an aliphatic group is a hydrocarbon moiety that may be straight chain or branched and may be completely saturated, or contain one or more units of unsaturation, but which is not aromatic. The term "unsaturated" means a moiety that has one or more double and/or triple bonds. The term "aliphatic" is therefore intended to encompass alkyl, alkenyl or alkynyl groups including multivalent equivalents such as alkylene, alkenylene and alkynylene, and combinations thereof. An aliphatic group is preferably a C_{1-20} aliphatic group, that is, an aliphatic group with 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19 or 20 carbon atoms. Preferably, an aliphatic group is a C_{1-15} aliphatic, more preferably a C_{1-12} aliphatic, more preferably a C_{1-10} aliphatic, even more preferably a C_{1-8} aliphatic, such as a C_{1-6} aliphatic group.

An alkyl group is preferably a " C_{1-20} alkyl group", that is an alkyl group that is a straight or branched chain with 1 to 20 carbons. The alkyl group therefore has 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19 or 20 carbon atoms. Preferably, an alkyl group is a C_{1-15} alkyl, preferably a C_{1-12} alkyl, more preferably a C_{1-10} alkyl, even more preferably a C_{1-8} alkyl, even more preferably a C_{1-6} alkyl group. Specifically, examples of " C_{1-20} alkyl group" include methyl group, ethyl group, n-propyl group, iso-propyl group, n-butyl group, iso-butyl group, sec-butyl group, tert-butyl group, n-pentyl group, n-hexyl group, n-heptyl group, n-octyl group, n-nonyl group, n-decyl group, n-undecyl group, n-dodecyl group, n-tridecyl group, n-tetradecyl group, n-pentadecyl group, n-hexadecyl group, n-heptadecyl group, n-octadecyl group, n-nonadecyl group, n-eicosyl group, 1,1-dimethylpropyl group, 1,2-dimethylpropyl group, 2,2-dimethylpropyl group, 1-ethylpropyl group, n-hexyl group, 1-ethyl-2-methylpropyl group, 1,1,2-trimethylpropyl group, 1-ethylbutyl group, 1-methylbutyl group, 2-methylbutyl group, 1,1-dimethylbutyl group, 1,2-dimethylbutyl group, 2,2-dimethylbutyl group, 1,3-dimethylbutyl group, 2,3-dimethylbutyl group, 2-ethylbutyl group, 2-methylpentyl group, 3-methylpentyl group and the like.

Alkenyl and alkynyl groups are preferably " C_{2-20} alkenyl" and " C_{2-20} alkynyl", more preferably " C_{2-15} alkenyl" and " C_{2-15} alkynyl", even more preferably " C_{2-12} alkenyl" and " C_{2-12} alkynyl", even more preferably " C_{2-10} alkenyl" and " C_{2-10} alkynyl", even more preferably " C_{2-8} alkenyl" and " C_{2-8} alkynyl", most preferably " C_{2-6} alkenyl" and " C_{2-6} alkynyl" groups, respectively.

Alkylene is divalent but otherwise defined as an alkyl group above. Likewise, alkenylene and alkynylene are defined as divalent equivalents of alkenyl and alkynyl above.

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A heteroaliphatic group (including heteroalkyl, heteroalkenyl and heteroalkynyl) is an aliphatic group as described above, which additionally contains one or more heteroatoms. Heteroaliphatic groups therefore preferably contain from 2 to 21 atoms, preferably from 2 to 16 atoms, more preferably from 2 to 13 atoms, more preferably from 2 to 11 atoms, more preferably from 2 to 9 atoms, even more preferably from 2 to 7 atoms, wherein at least one atom is a carbon atom. Particularly preferred heteroatoms are selected from O, S, N, P and Si. When heteroaliphatic groups have two or more heteroatoms, the heteroatoms may be the same or different.

Heteroalkylene is divalent but otherwise defined as a heteroalkyl group above. Likewise, heteroalkenylene and heteroalkynylene are defined as divalent equivalents of heteroalkenyl and heteroalkynyl above. An alicyclic group is a saturated or partially unsaturated cyclic aliphatic monocyclic or polycyclic (including fused, bridging and spiro-fused) ring system which has from 3 to 20 carbon atoms, that is an alicyclic group with 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19 or 20 carbon atoms. Preferably, an alicyclic group has from 3 to 15, more preferably from 3 to 12, even more preferably from 3 to 10, even more preferably from 3 to 8 carbon atoms, even more preferably from 3 to 6 carbons atoms. The term "alicyclic" encompasses cycloalkyl, cycloalkenyl and cycloalkynyl groups. It will be appreciated that the alicyclic group may comprise an alicyclic ring bearing one or more linking or non-linking alkyl substituents, such as $-\text{CH}_2\text{-cyclohexyl}$.

Specifically, examples of the C_{3-20} cycloalkyl group include cyclopropyl, cyclobutyl, cyclopentyl, cyclohexyl, cycloheptyl, adamantyl and cyclooctyl.

A heteroalicyclic group is an alicyclic group as defined above which has, in addition to carbon atoms, one or more ring heteroatoms, which are preferably selected from O, S, N, P and Si. Heteroalicyclic groups preferably contain from one to four heteroatoms, which may be the same or different. Heteroalicyclic groups preferably contain from 5 to 20 atoms, more preferably from 5 to 14 atoms, even more preferably from 5 to 12 atoms.

An aryl group is a monocyclic or polycyclic ring system having from 5 to 20 carbon atoms. An aryl group is preferably a " C_{6-12} aryl group" and is an aryl group constituted by 6, 7, 8, 9, 10, 11 or 12 carbon atoms and includes condensed ring groups such as monocyclic ring group, or bicyclic ring group and the like. Specifically, examples of " C_{6-10} aryl group" include phenyl group, biphenyl group, indenyl group, naphthyl group or azulenyl group and the like. It should be noted that condensed rings such as indan and tetrahydro naphthalene are also included in the aryl group.

A heteroaryl group is an aryl group having, in addition to carbon atoms, from one to four ring heteroatoms which are preferably selected from O, S, N, P and Si. A heteroaryl group preferably has from 5 to 20, more preferably from 5 to 14 ring atoms. Specifically, examples of a heteroaryl group include pyridine, imidazole, methylimidazole and dimethylaminopyridine.

Examples of alicyclic, heteroalicyclic, aryl and heteroaryl groups include but are not limited to cyclohexyl, phenyl, acridine, benzimidazole, benzofuran, benzothiophene, benzoxazole, benzothiazole, carbazole, cinnoline, dioxin, dioxane, dioxolane, dithiane, dithiazine, dithiazole, dithiolane, furan, imidazole, imidazoline, imidazolidine, indole, indoline, indolizine, indazole, isoindole, isoquinoline, isoxazole, isothiazole, morpholine, naphthyridine, oxazole, oxadiazole, oxathiazole, oxathiazolidine, oxazine, oxadiazine,

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phenazine, phenothiazine, phenoxazine, phthalazine, piperazine, piperidine, pteridine, purine, pyran, pyrazine, pyrazole, pyrazoline, pyrazolidine, pyridazine, pyridine, pyrimidine, pyrrole, pyrrolidine, pyrroline, quinoxaline, quinazoline, quinolizine, tetrahydrofuran, tetrazine, tetrazole, thiophene, thiadiazine, thiadiazole, thiazotriazole, thiazine, thiazole, thiomorpholine, thianaphthalene, thiopyran, triazine, triazole, and trithiane.

Arylene is divalent but otherwise defined as an aryl group above. Likewise heteroarylene is defined as divalent equivalents of heteroaryl and cycloalkylene as divalent equivalents of alicyclic and heteroalicyclic above.

The term "halide" or "halogen" or "halo" are used interchangeably and, as used herein mean a fluorine atom, a chlorine atom, a bromine atom, an iodine atom and the like, preferably a fluorine atom, a bromine atom or a chlorine atom, and more preferably a fluorine atom.

A haloalkyl group is preferably a " C_{1-20} haloalkyl group", more preferably a " C_{1-15} haloalkyl group", more preferably a " C_{1-12} haloalkyl group", more preferably a " C_{1-10} haloalkyl group", even more preferably a " C_{1-8} haloalkyl group", even more preferably a " C_{1-6} haloalkyl group" and is a C_{1-20} alkyl, a C_{1-15} alkyl, a C_{1-12} alkyl, a C_{1-10} alkyl, a C_{1-8} alkyl, or a C_{1-6} alkyl group, respectively, as described above substituted with at least one halogen atom, preferably 1, 2 or 3 halogen atom(s). Specifically, examples of " C_{1-20} haloalkyl group" include fluoromethyl group, difluoromethyl group, trifluoromethyl group, fluoroethyl group, difluoroethyl group, trifluoroethyl group, chloromethyl group, bromomethyl group, iodomethyl group and the like.

An alkoxy group is preferably a " C_{1-20} alkoxy group", more preferably a " C_{1-15} alkoxy group", more preferably a " C_{1-12} alkoxy group", more preferably a " C_{1-10} alkoxy group", even more preferably a " C_{1-8} alkoxy group", even more preferably a " C_{1-6} alkoxy group" and is an oxy group that is bonded to the previously defined C_{1-20} alkyl, C_{1-15} alkyl, C_{1-12} alkyl, C_{1-10} alkyl, C_{1-8} alkyl, or C_{1-6} alkyl group respectively. Specifically, examples of " C_{1-20} alkoxy group" include methoxy group, ethoxy group, n-propoxy group, iso-propoxy group, n-butoxy group, iso-butoxy group, sec-butoxy group, tert-butoxy group, n-pentyloxy group, isopentyloxy group, sec-pentyloxy group, n-hexyloxy group, iso-hexyloxy group, n-hexyloxy group, n-heptyloxy group, n-octyloxy group, n-nonyloxy group, n-decyloxy group, n-undecyloxy group, n-dodecyloxy group, n-tridecyloxy group, n-tetradecyloxy group, n-pentadecyloxy group, n-hexadecyloxy group, n-heptadecyloxy group, n-octadecyloxy group, n-nonadecyloxy group, n-eicosyloxy group, 1,1-dimethylpropoxy group, 1,2-dimethylpropoxy group, 2,2-dimethylpropoxy group, 2-methylbutoxy group, 1-ethyl-2-methylpropoxy group, 1,1,2-trimethylpropoxy group, 1,1-dimethylbutoxy group, 1,2-dimethylbutoxy group, 2,2-dimethylbutoxy group, 2,3-dimethylbutoxy group, 1,3-dimethylbutoxy group, 2-ethylbutoxy group, 2-methylpentyloxy group, 3-methylpentyloxy group and the like.

An aryloxy group is preferably a " C_{5-20} aryloxy group", more preferably a " C_{6-12} aryloxy group", even more preferably a " C_{6-10} aryloxy group" and is an oxy group that is bonded to the previously defined C_{5-20} aryl, C_{6-12} aryl, or C_{6-10} aryl group respectively.

An alkylthio group is preferably a " C_{1-20} alkylthio group", more preferably a " C_{1-15} alkylthio group", more preferably a " C_{1-12} alkylthio group", more preferably a " C_{1-10} alkylthio group", even more preferably a " C_{1-8} alkylthio group", even more preferably a " C_{1-6} alkylthio group" and is a thio ($-\text{S}-$) group that is bonded to the previously defined C_{1-20}

alkyl, C₁₋₁₅ alkyl, C₁₋₁₂ alkyl, C₁₋₁₀ alkyl, C₁₋₈ alkyl, or C₁₋₆ alkyl group respectively. An alkylthio group utilised as a substituent as defined herein, may be connected via either a carbon atom of the alkyl group as defined above or the sulphur atom of the thio group. An arylthio group is preferably a "C₅₋₂₀ arylthio group", more preferably a "C₆₋₁₂ arylthio group", even more preferably a "C₆₋₁₀ arylthio group" and is a thio (—S—) group that is bonded to the previously defined C₅₋₂₀ aryl, C₆₋₁₂ aryl, or C₆₋₁₀ aryl group respectively.

An alkylaryl group is preferably a "C₆₋₁₂ aryl C₁₋₂₀ alkyl group", more preferably a "C₆₋₁₂ aryl C₁₋₁₆ alkyl group", even more preferably a "C₆₋₁₂ aryl C₁₋₆ alkyl group" and is an aryl group as defined above bonded at any position to an alkyl group as defined above. The point of attachment of the alkylaryl group to a molecule may be via the alkyl portion and thus, preferably, the alkylaryl group is —CH₂—Ph or —CH₂CH₂—Ph. An alkylaryl group can also be referred to as "aralkyl".

A silyl group is preferably a group —Si(R_s)₃, wherein each R_s can be independently an aliphatic, heteroaliphatic, alicyclic, heteroalicyclic, aryl or heteroaryl group as defined above. In certain embodiments, each R_s is independently an unsubstituted aliphatic, alicyclic or aryl. Preferably, each R_s is an alkyl group selected from methyl, ethyl or propyl.

A silyl ether group is preferably a group OSi(R_s)₃ wherein each R_s can be independently an aliphatic, heteroaliphatic, alicyclic, heteroalicyclic, aryl or heteroaryl group as defined above. In certain embodiments, each R_s can be independently an unsubstituted aliphatic, alicyclic or aryl. Preferably, each R_s is an optionally substituted phenyl or optionally substituted alkyl group selected from methyl, ethyl, propyl or butyl (such as n-butyl or tert-butyl (tBu)). Exemplary silyl ether groups include OSi(CH₃)₃, OSi(C₂H₅)₃, OSi(C₆H₅)₃, OSi(CH₃)₂(tBu), OSi(tBu)₃ and OSi(C₆H₅)₂(tBu).

A nitrile group (also referred to as a cyano group) is a group CN.

An imine group is a group —CR₆NR₆, preferably a group —CHNR₆ wherein R₆ is an aliphatic, heteroaliphatic, alicyclic, heteroalicyclic, aryl or heteroaryl group as defined above. In certain embodiments, R₆ is unsubstituted aliphatic, alicyclic or aryl. Preferably R₆ is an alkyl group selected from methyl, ethyl or propyl.

An acetylide group contains a triple bond —C≡C—R₇, preferably wherein R₇ can be hydrogen, an aliphatic, heteroaliphatic, alicyclic, heteroalicyclic, aryl or heteroaryl group as defined above. For the purposes of the invention when R₇ is alkyl, the triple bond can be present at any position along the alkyl chain. In certain embodiments, R₇ is unsubstituted aliphatic, alicyclic or aryl. Preferably R₇ is methyl, ethyl, propyl or phenyl.

An amino group is preferably —NH₂, —NHR₈ or —N(R₈)₂ wherein R₈ can be an aliphatic, heteroaliphatic, alicyclic, heteroalicyclic, a silyl group, aryl or heteroaryl group as defined above. It will be appreciated that when the amino group is N(R₈)₂, each R₈ group can be the same or different. In certain embodiments, each R₈ is independently an unsubstituted aliphatic, alicyclic, silyl or aryl. Preferably R₈ is methyl, ethyl, propyl, Si(CH₃)₃ or phenyl.

An amido group is preferably —NR₉C(O)R₉ or —C(O)—NR₉(R₉) wherein R₉ can be hydrogen, an aliphatic, heteroaliphatic, alicyclic, heteroalicyclic, aryl or heteroaryl group as defined above. In certain embodiments, R₉ is unsubstituted aliphatic, alicyclic or aryl. Preferably R₉ is hydrogen, methyl, ethyl, propyl or phenyl. The amido group may be

terminated by hydrogen, an aliphatic, heteroaliphatic, alicyclic, heteroalicyclic, aryl or heteroaryl group.

An ester group is preferably —OC(O)R₁₀ or —C(O)OR₁₀ wherein R₁₀ can be hydrogen, an aliphatic, heteroaliphatic, alicyclic, heteroalicyclic, aryl or heteroaryl group as defined above. In certain embodiments, R₁₀ is unsubstituted aliphatic, alicyclic or aryl. Preferably R₁₀ is hydrogen, methyl, ethyl, propyl or phenyl. The ester group may be terminated by hydrogen, an aliphatic, heteroaliphatic, alicyclic, heteroalicyclic, aryl or heteroaryl group.

An ether group is preferably —OR₁₅ or —R₁₆OR₁₇ wherein R₁₅, R₁₆ and R₁₇ can be an aliphatic, heteroaliphatic, alicyclic, heteroalicyclic, aryl or heteroaryl group as defined above. In certain embodiments, R₁₅, R₁₆ and R₁₇ are each unsubstituted aliphatic, alicyclic or aryl. Preferably, R₁₅, R₁₆ and R₁₇ are each methyl, ethyl, propyl or phenyl. The ether group may be terminated by hydrogen, an aliphatic, heteroaliphatic, alicyclic, heteroalicyclic, aryl or heteroaryl group.

A sulfoxide is preferably —S(O)R₁₁ and a sulfonyl group is preferably —S(O)₂R₁₁ wherein R₁₁ can be hydrogen, an aliphatic, heteroaliphatic, alicyclic, heteroalicyclic, aryl or heteroaryl group as defined above. In certain embodiments, R₁₁ is unsubstituted aliphatic, alicyclic or aryl. Preferably R₁₁ is hydrogen, methyl, ethyl, propyl or phenyl.

A sulfinate group is preferably —OSOR₁₂ wherein R₁₂ can be hydrogen, an aliphatic, heteroaliphatic, haloaliphatic, alicyclic, heteroalicyclic, aryl or heteroaryl group as defined above. In certain embodiments, R₁₂ is unsubstituted aliphatic, alicyclic or aryl. Preferably R₁₂ is hydrogen, methyl, ethyl, propyl or phenyl.

Groups R₁₃, R₁₄, R₁₈, R₁₉, R₂₀ and R₂₁ can be a hydrogen an aliphatic, heteroaliphatic, alicyclic, heteroalicyclic, aryl or heteroaryl group as defined above. In certain embodiments, R₁₃, R₁₄, R₁₈, R₁₉, R₂₀ and R₂₁ are each unsubstituted aliphatic, alicyclic or aryl. Preferably, R₁₃, R₁₄, R₁₈, R₁₉, R₂₀ and R₂₁ are each hydrogen, methyl, ethyl, propyl or phenyl.

By the term "phosphonium" as used herein is meant the cation comprising the formula PH₄⁺.

Any of the aliphatic (including alkyl, alkenyl, alkynyl, alkylenes, alkenylenes and alkynylenes), heteroaliphatic, (including heteroalkyl, heteroalkenyl, heteroalkynyl, heteroalkylene, heteroalkenylenes and heteroalkynylenes), alicyclic, carboxylic, carboxylate, cycloalkylene, heteroalicyclic, aryl, arylenes, heteroaryl, heteroarylenes, haloalkyl, alkoxy, aryloxy, alkylthio, arylthio, alkylaryl, silyl, silyl ether, ester, sulfoxide, sulfonyl, imine, acetylide, amino, sulfonate or amido groups wherever mentioned above, particularly when mentioned as optionally substituted above may be optionally substituted by halogen, hydroxy, nitro, alkoxy, aryloxy, alkylthio, arylthio, heteroaryloxy, alkylaryl, amino, amido, imine, nitrile, silyl, silyl ether, ester, sulfoxide, sulfonyl, acetylide, sulfonate or optionally substituted aliphatic, heteroaliphatic, alicyclic, heteroalicyclic, aryl or heteroaryl groups (for example, optionally substituted by halogen, hydroxy, nitro, alkoxy, aryloxy, alkylthio, arylthio, amino, imine, nitrile, silyl, sulfoxide, sulfonyl, sulfonate or acetylide).

When we use the term "optionally substituted" at the start of a list of chemical species we mean that all of the species in the list which can be substituted may be optionally substituted; that is we do not mean that only the first species mentioned in the list may be optionally substituted.

All of the features contained herein may be combined with any of the above aspects and in any combination.

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Embodiments of the invention will now be described with reference to the following non-limiting examples.

Non-alkylated biphenol ligand precursors used in the following examples are described in WO2016/012785.

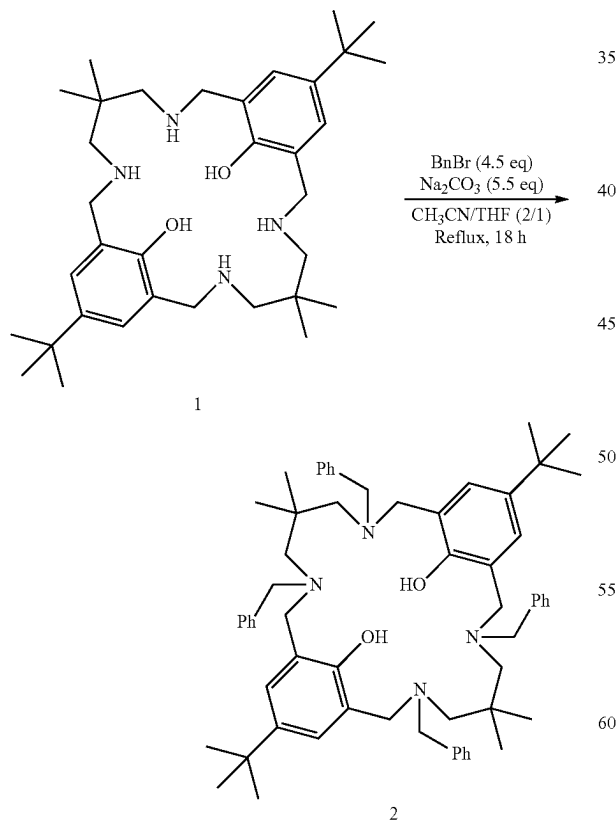
Partially alkylated biphenol ligand precursors used in the following examples were prepared using protecting group chemistry in the absence of a base, as described in WO2016/012785.

The invention will be further illustrated by means of the following non-binding examples.

Experimental Description

It should be noted here that the examples given demonstrate the breadth of applicability of the process of the present invention. The yields reported in the following examples are substantially unoptimized and reaction conditions have not been extensively tailored to suit individual target macrocycles. Some of the yields are therefore <40%, but it is understood with variation of the conditions that these yields could be substantially improved. In many cases very high yields were obtained, even without optimisation. It is noted that in each case, and as demonstrated by the enclosed NMR data, no alkylation of the phenols was observed. The products obtained were the desired tetra-N-alkylated-diphenol compounds.

Example Set 1: Preparation of Tetra-benzylated Ligand 2

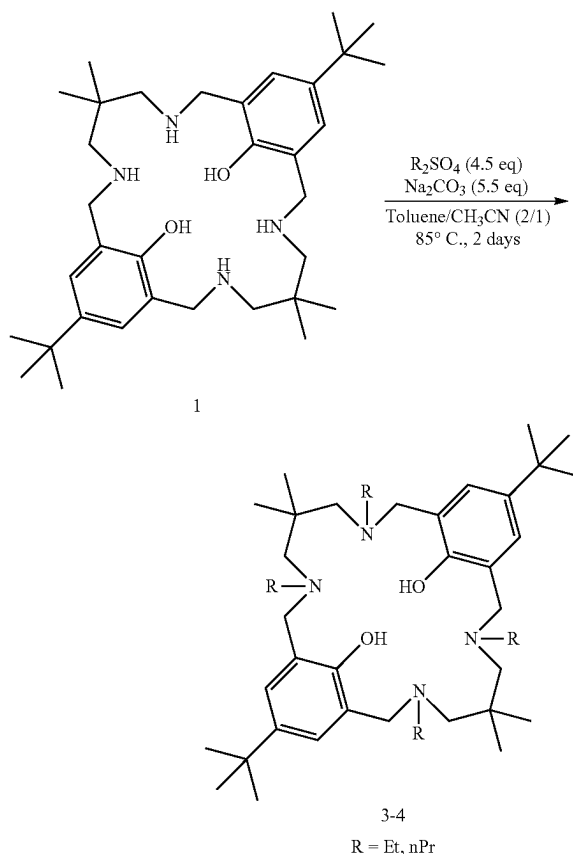


To a solution of ligand 1 (1.00 g, 1.0 eq) in tetrahydrofuran (10 mL) and acetonitrile (20 mL) were added benzyl

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bromide (0.96 mL, 4.5 eq) and sodium carbonate (1.05 g, 5.5 eq). The reaction mixture was stirred at reflux for 18 h. After this time, reaction mixture was filtered and the filter cake was washed with acetonitrile. The filter cake was solubilised in dichloromethane/water, and phases were separated. The aqueous phase was extracted with dichloromethane. The combined organic layers were dried over sodium sulfate, and the solvents were evaporated in vacuo to yield tetra-benzylated ligand 2 (>85%). MS (ESI): 913.6 [M+H]⁺; ¹H NMR (400 MHz, CDCl₃) δ (ppm) 10.60 (s, 2H, phenols), 7.45-7.41 (m, 8H), 7.39-7.34 (m, 8H), 7.31-7.26 (m, 4H), 6.89 (s, 4H), 3.75-3.64 (m, 16H), 2.54 (s, 8H), 1.26 (s, 18H), 0.85 (s, 12H).

Example Set 2: Preparation of Tetra-alkylated Ligands 3, 4

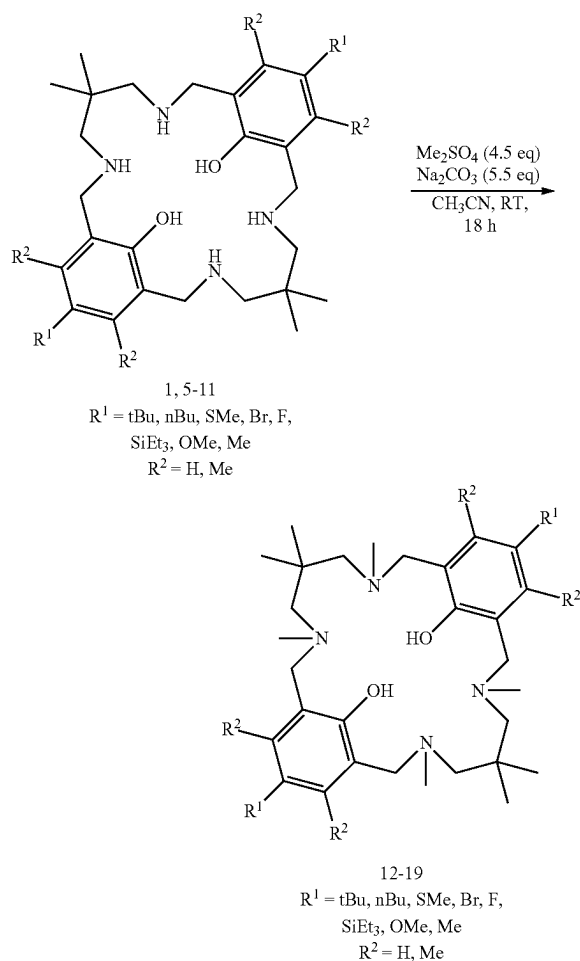


Representative procedure: To a solution of ligand 1 (1.00 g, 1.0 eq) in toluene (12 mL) and acetonitrile (24 mL) were added diethylsulfate (1.06 mL, 4.5 eq) and sodium carbonate (1.05 g, 5.5 eq). Reaction mixture was stirred at 85° C. for 2 days. After this time, reaction mixture was filtered and the mother liquor was evaporated. The residue was solubilised in dichloromethane/water, and phases were separated. The aqueous phase was extracted with dichloromethane. The combined organic layers were washed with a saturated aqueous solution of sodium hydrogen carbonate, and phases were separated. The organic layer was dried over sodium sulfate, and the solvents were evaporated in vacuo to yield tetra-alkylated ligand 3 (33%). ¹H NMR (400 MHz, CDCl₃) δ (ppm) 10.68 (s, 2H, phenols), 6.95 (s, 4H), 3.63 (s, 8H),

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2.55 (q, J=6.9 Hz, 8H), 2.37 (s, 8H), 1.26 (s, 18H), 1.07 (t, J=6.9 Hz, 12H), 0.78 (s, 12H). MS (ESI): 665.6 [M+H]⁺
Mass spectrometry data for compound 4 (28%): MS (ESI): 721.6 [M+H]⁺

Example Set 3: Preparation of Tetra-methylated
Ligands 12-19



Representative procedure: To a suspension of ligand 1 (R¹=tBu, R²=H) (1.00 g, 1.0 eq) in acetonitrile (30 mL) were added Me₂SO₄ (0.77 mL, 4.5 eq) and sodium carbonate (1.05 g, 5.5 eq). Reaction mixture was stirred at room temperature (RT) for 18 h. After this time, reaction mixture was filtered and the filter cake was washed with acetonitrile. The filter cake was solubilised in dichloromethane/water, and phases were separated. The aqueous phase was extracted with dichloromethane. The combined organic layers were dried over sodium sulfate, and the solvents were evaporated in vacuo to yield tetra-methylated ligand 12 (R¹=tBu, R²=H>70%) as a white powder. ¹H NMR (400 MHz, CDCl₃) δ (ppm) 10.99 (s, 2H, phenols), 6.96 (s, 4H), 3.59 (s, 8H), 2.38 (s, 8H), 2.33 (s, 12H), 1.29 (s, 18H), 0.89 (s, 12H). MS (ESI) 609.4 [M+H]⁺

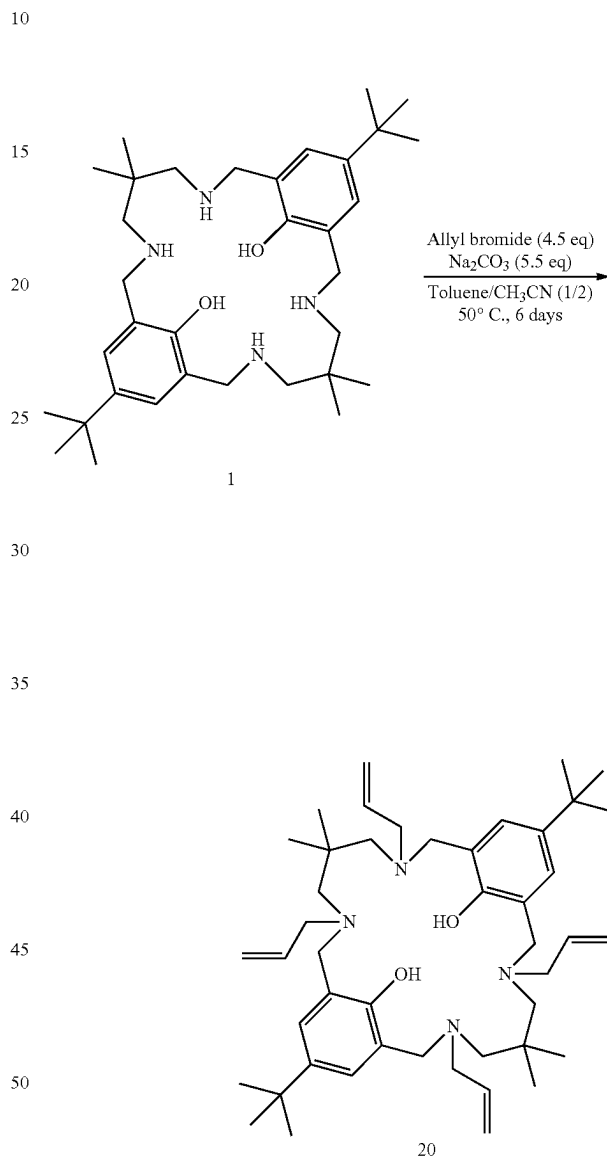
Mass spectrometry data for compounds 13-19:

13 (56%) (R¹=SMe, R²=H): MS (ESI) 589.3 [M+H]⁺
14 (53%) (R¹=Br, R²=H): MS (ESI) 655.2 [M+H]⁺
15 (52%) (R¹=F, R²=H): MS (ESI) 533.4 [M+H]⁺

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16 (59%) (R¹=SiEt₃, R²=H): MS (ESI) 725.5 [M+H]⁺
17 (33%) (R¹=OMe, R²=H): MS (ESI) 557.4 [M+H]⁺
18 (40%) (R¹=nBu, R²=H): MS (ESI) 609.4 [M+H]⁺
19 (40%) (R¹=Me, R²=Me): MS (ESI) 581.4 [M+H]⁺

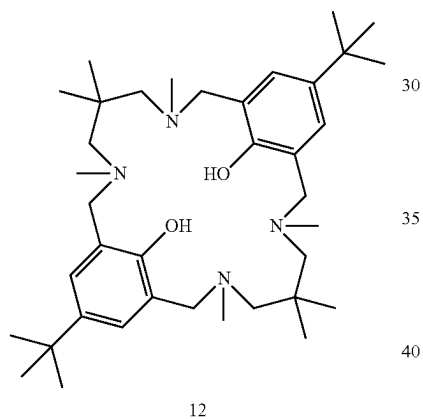
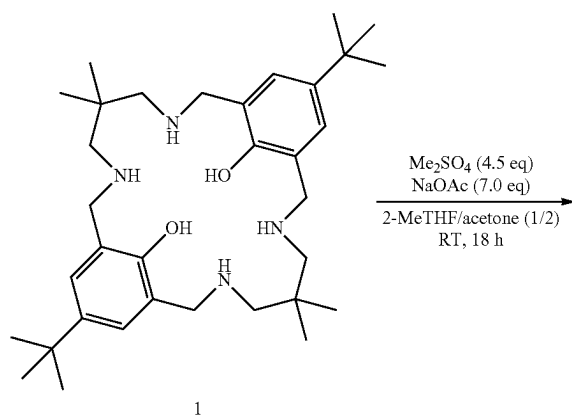
Example 4: Preparation of Tetra-allylated Ligand
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To a solution of ligand 1 (1.00 g, 1.0 eq) in toluene (10 mL) and acetonitrile (20 mL) were added allyl bromide (0.70 mL, 4.5 eq) and sodium carbonate (1.05 g, 5.5 eq). Reaction mixture was stirred at 50° C. for 6 days. After this time, reaction mixture was filtered and the filter cake was washed with acetonitrile. The filter cake was solubilised in dichloromethane/water, and phases were separated. The aqueous phase was extracted with dichloromethane. The combined organic layers were dried over sodium sulfate, and the solvents were evaporated in vacuo to yield tetra-allylated ligand 20 (>90%) as a white powder. MS (ESI) 713.6 [M+H]⁺

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Example 5: Further Preparation of Tetra-methylated Ligand 12



Representative example: To a solution of ligand 1 (1.00 g, 1.0 eq) in 2-methyl-tetrahydrofuran (18 mL) and acetone (18 mL) were added dimethylsulfate (0.77 mL, 4.5 eq) and sodium ethanoate (1.04 g, 7.0 eq). Reaction mixture was stirred at room temperature for 18 h. After this time, reaction mixture was filtered and the filter cake was washed with acetonitrile. The filter cake was solubilised in dichloromethane/water, and phases were separated. The aqueous phase was extracted with dichloromethane. The combined organic layers were dried over sodium sulfate, and the solvents were evaporated in vacuo to yield tetra-methylated ligand 12 (35%).

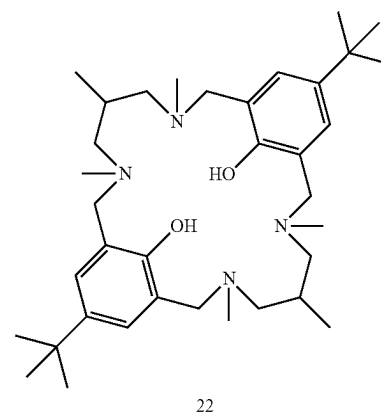
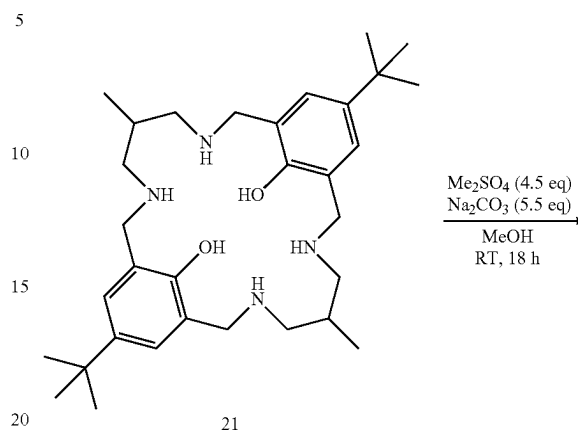
Examples of other bases used in the preparation of ligand 12:

Base=sodium hydroxide (NaOH—5.5 eq),
solvent=tetrahydrofuran/acetonitrile (1:3), room temperature, 16 hours. Yield=18%.

Base=disodium phosphate (Na₂HPO₄—5.5 eq),
solvent=tetrahydrofuran/acetonitrile (1:3), room temperature, 16 hours. Yield=20%.

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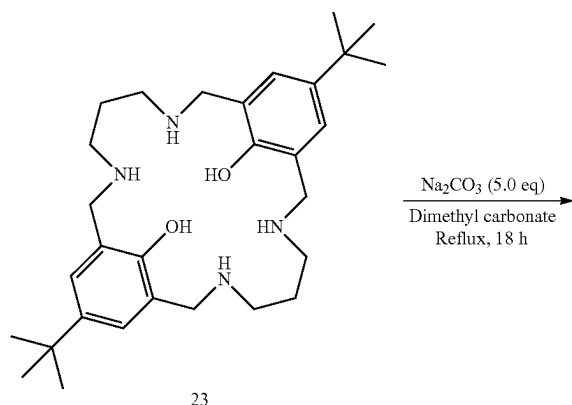
Example 6: Preparation of Tetra-methylated Ligand 22



To a solution of ligand 21 (1.00 g, 1.0 eq) in methanol (30 mL) were added dimethylsulfate (0.81 mL, 4.5 eq) and sodium carbonate (1.10 g, 5.5 eq). Reaction mixture was stirred at room temperature for 18 h. After this time, reaction mixture was filtered and the mother liquor was evaporated. The residue was solubilised in dichloromethane/water, and phases were separated. The aqueous phase was extracted with dichloromethane. The combined organic layers were washed with a saturated aqueous solution of sodium hydrogen carbonate, and phases were separated. The organic layer was dried over sodium sulfate, and the solvents were evaporated. 22 (60%); MS (ESI) 581.4 [M+H]⁺

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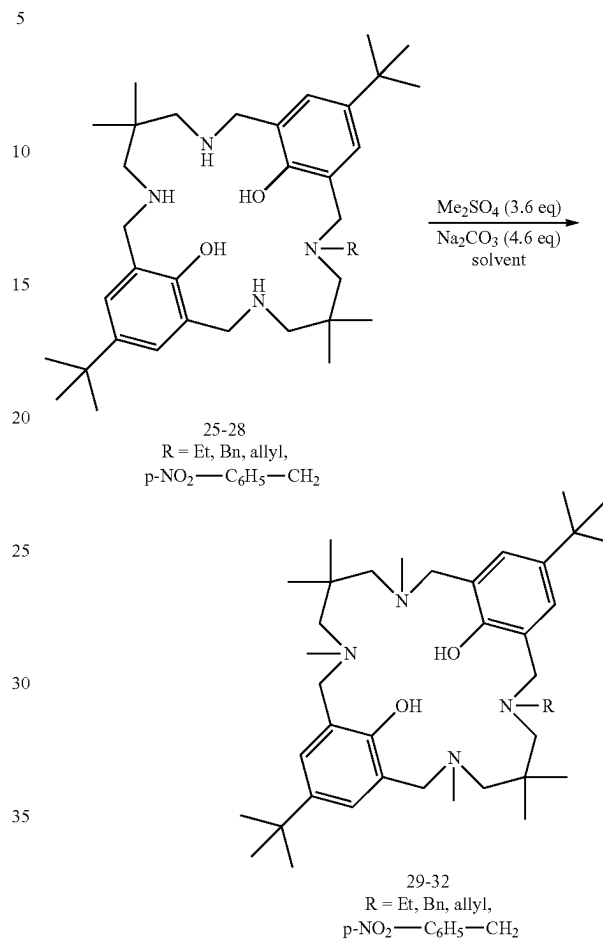
Example 7: Preparation of Tetra-methylated Ligand 24



To a solution of ligand 23 (1.00 g, 1.0 eq) in dimethyl carbonate (30 mL) was added sodium carbonate (1.06 g, 5.0 eq). Reaction mixture was stirred at reflux for 18 h. After this time, reaction mixture was filtered and the mother liquor was evaporated. The residue was solubilised in dichloromethane/water, and phases were separated. The aqueous phase was extracted with dichloromethane. The combined organic layers were washed with a saturated aqueous solution of sodium hydrogen carbonate, and phases were separated. The organic layer was dried over sodium sulfate, and the solvents were evaporated. 24 (45%): MS (ESI) 553.4 [M+H]⁺

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Example 8: Preparation of Tri-methylated Ligands 29-32



Representative procedure: To a solution of 25 (0.835 g, 1.4 mmol) in THF (5 mL) and MeCN (10 mL) was added Me₂SO₄ (0.48 mL, 5.0 mmol), followed by Na₂CO₃ (0.690 g, 6.5 mmol). The reaction mixture was stirred at room temperature of 18 h. After this time, reaction mixture was filtered and the filter cake was washed with acetonitrile. The solid collected was suspended in DCM, washed with H₂O and saturated NaHCO₃, before being dried over MgSO₄ and reduced in volume to give 29 (70% yield) as a white solid. ¹H NMR (400 MHz, CDCl₃) δ (ppm) 10.92 (br s, 2H, phenol), 7.00 (d, 1H, J=2.5 Hz), 6.99 (d, 1H, J=2.5 Hz), 6.85 (d, 1H, J=2.5 Hz), 6.84 (d, 1H, J=2.5 Hz), 3.65 (s, 2H), 3.62 (s, 2H), 3.42 (s, 2H), 3.41 (s, 2H), 2.56 (q, 4H, J=7.0 Hz), 2.40 (s, 2H), 2.37 (s, 2H), 2.33 (s, 3H), 2.33 (s, 2H), 2.31 (s, 2H), 2.25 (s, 3H), 2.23 (s, 3H), 1.25 (2xs, 18H), 1.05 (t, 3H, J=7.0 Hz), 0.86 (s, 6H), 0.83 (s, 6H). MS (ESI) m/z=623.5 [M+H]⁺.

Data for compounds 30-32:

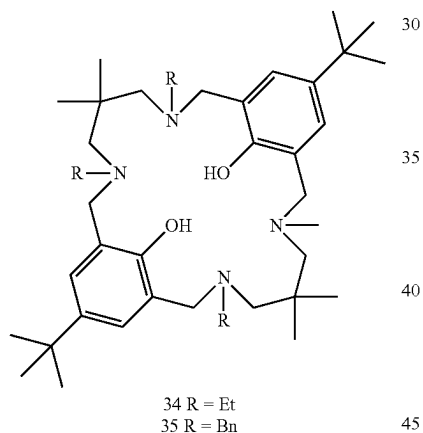
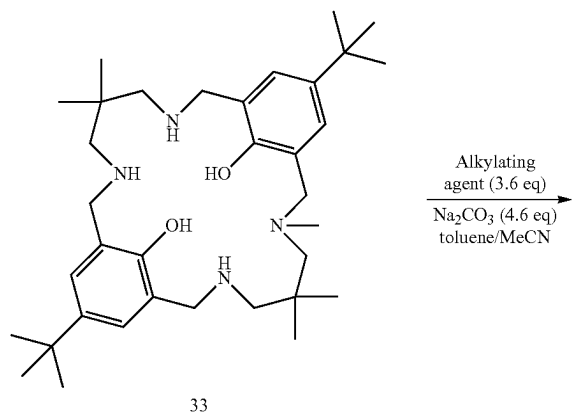
30 (84%) (R=Bn): MS (ESI) 685.4 [M+H]⁺

31 (67%) (R=allyl): MS (ESI) 635.5 [M+H]⁺

32 (73%) (R=p-NO₂-C₆H₅-CH₂): MS (ESI) 730.5 [M+H]⁺

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Example 9: Preparation of Tri-ethylated and Tribenzylated Ligands 34, 35



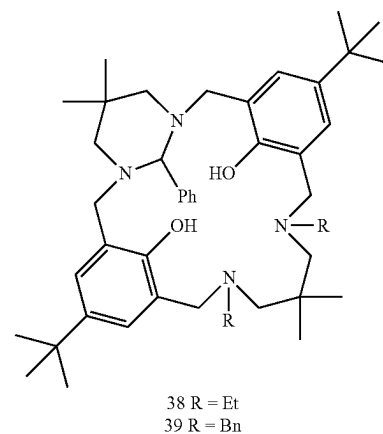
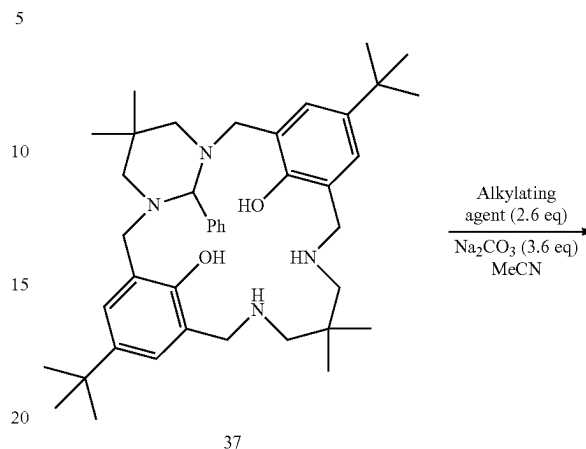
Representative example: To a solution of 33 (1.501 g, 2.6 mmol) in toluene (5 mL) and MeCN (8 mL) was added Et₂SO₄ (1.2 mL, 9.3 mmol), followed by Na₂CO₃ (1.267 g, 11.9 mmol). The reaction was stirred at 60° C. overnight. The filter cake was collected by filtration and washed with MeCN. The solid collected was suspended in DCM, washed with H₂O and saturated NaHCO₃, before being dried over MgSO₄ and reduced in volume to give 34 (65% yield) as a white solid.

¹H NMR (400 MHz, CDCl₃) δ (ppm) 10.80 (br s, 2H, phenol), 6.97 (dd, 4H, J=2.5, 11.6 Hz), 6.86 (dd, 4H, J=2.4, 7.4 Hz), 3.66 (d, 4H, J=7.8 Hz), 3.50 (s, 2H), 3.46 (s, 2H), 2.53 (m, 6H), 2.39 (d, 4H, J=7.5 Hz), 2.35 (s, 2H), 2.31 (s, 2H), 2.23 (3H, s), 1.25 (18H, d), 1.05 (6H, td, J=1.5, 7.0 Hz), 1.01 (t, 3H, J=7.0 Hz), 0.82 (s, 6H), 0.79 (s, 6H). MS (ESI) 651.6 [M+H]⁺

Data for compound 35 (74%): As per the representative example, except BnBr was used as the alkylating agent and toluene was substituted for THF. MS (ESI) 837.6 [M+H]⁺

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Example 10: Preparation of Di-ethylated and Di-benzylated Ligands 38, 39

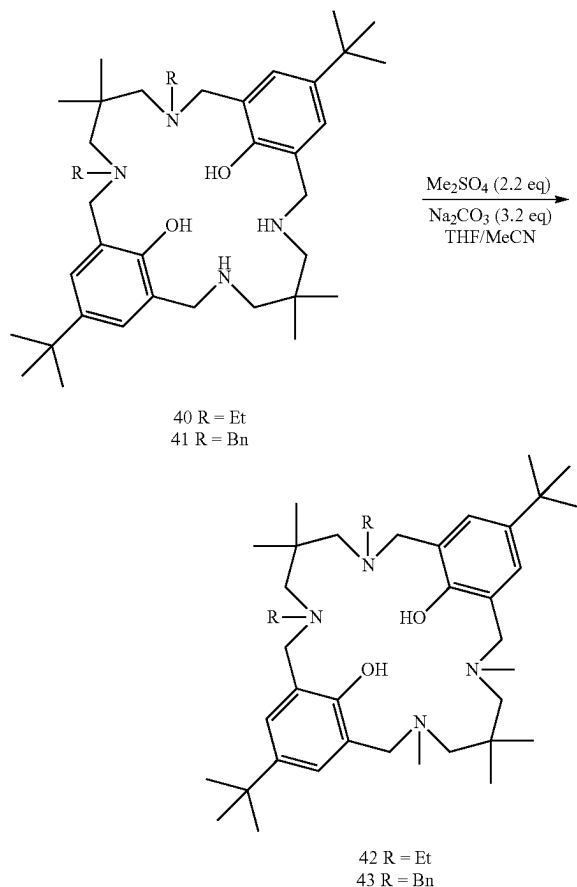


Representative example: 37 (2.000 g, 3 mmol), BnBr (0.93 mL, 8 mmol), and Na₂CO₃ (1.182 g, 11 mmol) were suspended in MeCN (30 mL) and stirred at 40° C. for 2 days. The reaction mixture was filtered, extracted into DCM, washed with NaHCO₃ (×3), dried over MgSO₄, and reduced in volume to yield 39 in 75% yield. MS (ESI) 821.6 [M+H]⁺

Data for compound 38 (80%): As per representative example except Et₂SO₄ was used as the alkylating agent and the reaction was carried out in THF/MeCN (1:2). MS (ESI) 697.5 [M+H]⁺.

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Example 11: Preparation of
Cis-dimethyl-dialkylated Ligands 42 and 43



Representative procedure: To a solution of 40 (1 g, 1.6 mmol) in THF (3 mL) and MeCN (8 mL) was added Me₂SO₄ (0.34 mL, 3.6 mmol), followed by Na₂CO₃ (0.56 g, 5.3 mmol). The reaction mixture was stirred at room temperature overnight. After this time, the reaction mixture was filtered and the filter cake was washed with acetonitrile. The solid collected was suspended in DCM, washed with H₂O and saturated NaHCO₃, before being dried over MgSO₄ and reduced in volume to give 42 (72% yield) as a white solid. MS (ESI) 637.2 [M+H]⁺.

Data for compound 43 (84%), MS (ESI) 761.2 [M+H]⁺.

Attention is directed to all papers and document which are filed concurrently with or previous to this specification in connection with this application and which are open to the public inspection with this specification, and the contents of all such papers and documents are incorporated herein by reference.

All of the features disclosed in this specification (including any accompanying claims, abstract and drawings), and/or all of the steps of any method or process so disclosed, may be combined in any combination, expect combinations where at least some of such features and/or steps are mutually exclusive.

Each feature disclosed in this specification (including any accompanying claims, abstract and drawings) may be replaced by alternative features serving the same, equivalent or similar purpose, unless expressly stated otherwise. Thus,

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unless expressly stated otherwise, each feature disclosed is one example only of a generic series of equivalent or similar features.

The invention is not restricted to the details of the foregoing embodiment(s). The invention extends to any novel one, or any novel combination, of the features disclosed in this specification (including any accompanying claims, abstract or drawings), or to any novel one, or any novel combinations, of the steps of any method or process so disclosed.

What is claimed is:

1. A compound of structure (Ie)

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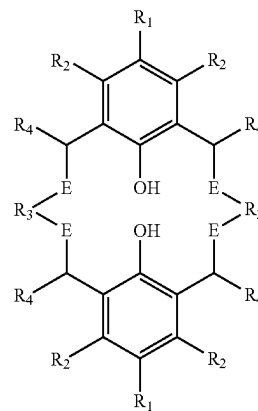
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(Ie)



wherein R₁ and R₂ are independently selected from hydrogen, halide, a nitro group, a nitrile group, an imine group, —NCR₁₃R₁₄, an amine, an ether group —OR₁₅ or —R₁₆OR₁₇, an ester group —OC(O)R₁₀ or —C(O)OR₁₀, an amido group —NR₉C(O)R₉ or —C(O)—NR₉(R₉), —COOH, —C(O)R₁₅, —OP(O)(OR₁₈)(OR₁₉), —P(O)R₂₀R₂₁, a silyl group, a silyl ether group, a sulfoxide group, a sulfonyl group, a sulfinate group or an acetylide group or an optionally substituted alkyl, alkenyl, alkynyl, haloalkyl, aryl, heteroaryl, alkoxy, aryloxy, alkylthio, arylthio, alicyclic or heteroalicyclic group;

R₃ is independently selected from optionally substituted alkylene, alkenylene, alkynylene, heteroalkylene, heteroalkenylene, heteroalkynylene, arylene, heteroarylene or cycloalkylene, wherein alkylene, alkenylene, alkynylene, heteroalkylene, heteroalkenylene and heteroalkynylene, in each case optionally interrupted by aryl, heteroaryl, alicyclic or heteroalicyclic; R₄ is independently selected from hydrogen, or optionally substituted aliphatic, heteroaliphatic, alicyclic, heteroalicyclic, aryl, heteroaryl, alkylheteroaryl or alkylaryl;

and E is independently selected from NR₅ and NR₆,

wherein R₅ is independently selected from optionally substituted aliphatic, heteroaliphatic, alicyclic, alkanolate, arylate, carboxyl, heteroalicyclic, aryl, heteroaryl, alkylheteroaryl or alkylaryl;

or two R₅ species may together be selected from optionally substituted alkylene, alkenylene or alkynylene, bonded to two different N groups of the compound of structure (Ie);

wherein R₆ is independently selected from optionally substituted aliphatic, alicyclic, heteroalicyclic, aryl,

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- heteroaryl, ether, polyether, or optionally substituted alkylaryl or alkylheteroaryl;
- with the first proviso that at least one of the species E conforms to the definition given for NR₆;
- and with the second proviso that the groups R₅ and R₆ are not all identical with each other;
- wherein R₉, R₁₀, R₁₃, R₁₄, R₁₈, R₁₉, R₂₀ and R₂₁ are independently selected from hydrogen or an optionally substituted aliphatic, heteroaliphatic, alicyclic, heteroalicyclic, aryl or heteroaryl group and R₁₅, R₁₆ and R₁₇ are independently selected from an optionally substituted aliphatic, heteroaliphatic, alicyclic, heteroalicyclic, aryl or heteroaryl group.
2. The compound according to claim 1, wherein each occurrence of R₁ and/or each occurrence of R₂, is the same.
3. The compound according to claim 1, wherein R₁ is different from R₂.
4. The compound according to claim 1, wherein R₁ and R₂ are independently selected from hydrogen, halide, amino, nitro, sulfoxide, sulfonyl, sulfinate, silyl, silyl ether and an optionally substituted alkyl, alkenyl, aryl, heteroaryl, heteroalicyclic, alkoxy, aryloxy or alkylthio or arylthio.
5. The compound according to claim 4, wherein R₂ is hydrogen or alkyl and R₁ is independently selected from hydrogen, C₁₋₆alkyl, C₁₋₆haloalkyl, alkoxy, aryl, halide, nitro, sulfonyl, silyl or alkylthio.
6. The compound according to claim 5, wherein R₂ is hydrogen and R₁ is independently selected from t-butyl, n-butyl, i-propyl, methyl, piperidinyl, methoxy, hexyl methyl ether, —SCH₃, —S(C₆H₅), H, nitro, trimethylsilyl, triethylsilyl, methylsulfonyl (—SO₂CH₃), triethylsilyl, halogen or phenyl.
7. The compound according to claim 1, wherein R₃ is independently selected from alkylene, alkenylene, alkynylene, heteroalkylene, heteroalkenylene or heteroalkynylene group which may optionally be interrupted by an aryl, heteroaryl, alicyclic or heteroalicyclic group, or may be a divalent arylene or cycloalkylene group which acts as a bridging group between two nitrogen centres in the compound of structure (Ie).
8. The compound according to claim 7, wherein R₃ is independently selected from a substituted or unsubstituted alkylene, cycloalkylene, alkenylene, heteroalkylene and arylene.
9. The compound according to claim 8, wherein R₃ is independently selected from 2,2-dimethylpropane-1,3-diyl, ethane-1,2-diyl, 2,2-fluoropropane-1,3-diyl, propane-1,3-

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- diyl, butane-1,4-diyl, phenylene, cyclohexane-1,4-diyl, cyclohexane-1,2-diyl or biphenylene.
10. The compound according to claim 9, wherein R₃ is independently selected from 2,2-dimethylpropane-1,3-diyl, ethane-1,2-diyl, 2,2-fluoropropane-1,3-diyl, phenylene, cyclohexane-1,4-diyl, cyclohexane-1,2-diyl or biphenylene.
11. The compound according to claim 1, wherein R₄ is independently selected from hydrogen, and optionally substituted aliphatic or aryl.
12. The compound according to claim 11, wherein R₄ is independently selected from hydrogen, and optionally substituted alkyl or aryl.
13. The compound according to claim 12, wherein R₄ is independently selected from hydrogen, methyl, ethyl, n-propyl, n-butyl, phenyl and trifluoromethyl.
14. The compound according to claim 1, wherein R₅ is independently selected from optionally substituted alkyl which is optionally interrupted by at least one N atom, or by at least one O atom, or by at least one S atom, optionally substituted alkylthio, alkylaryl, alkylheteroaryl, alkenyl, alkynyl, heteroalkenyl, heteroalkynyl, aryl, alicyclic, heteroalicyclic, heteroaryl, sulfonate, alkanoate (—C(O)—OR₁₀), arylate (arylC(O)O—), aryl-C(O)OR₁₀, alkylaryl-C(O)OR₁₀, alkyl-C(O)—OR₁₀, carbonyl (—C(O)—R₁₀), ether, polyether, alkylarylC(O)O— or carboxyl; or two R₅ groups together comprise an optionally substituted alkylene group bonded to two different N groups on the same macrocycle.
15. The compound according to claim 14, wherein R₅ is independently selected from optionally substituted alkyl, alkylaryl, alkenyl, alkynyl, sulfonate, alkanoate, aryl-C(O)OR₁₀, alkylaryl-C(O)OR₁₀, alkyl-C(O)—OR₁₀ or alkyl-C≡N.
16. The compound according to claim 15, wherein R₅ is independently selected from methyl, ethyl, propyl, butyl, allyl, propargyl, benzyl, 4-nitrobenzyl, —CH₂CH₂—C(O)—OR₁₀, —CH₂CH₂—C≡N, or —C(1-4)alkylC(O)O—, or benzoate which is unsubstituted or ring-substituted by 1, 2 or 3 C(1-4)alkyl groups.
17. The compound according to claim 1, wherein R₆ is independently selected from optionally substituted alkyl, alicyclic, alkenyl, alkynyl, alkylaryl, methylpyridine, ether, polyethylene glycol, or polypropylene glycol.
18. The compound according to claim 17, wherein R₆ is independently selected from an optionally substituted alkylaryl, C(2-10)alkenyl group, C(2-10)alkynyl group, C(1-10)alkyl, allyl, propargyl or benzyl.

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